```
C:\WINDOWS\system32\cmd.   Windows PowerShell   Windows PowerShell

Microsoft Windows [Version 10.0.26200.8875]
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C:\Users\Sanjana CA>ollama --version
Warning: could not connect to a running Ollama instance
Warning: client version is 0.32.6

C:\Users\Sanjana CA>ollama serve
time=2026-08-10T16:14:38.697+05:30 level=INFO source=routes.go:1933 msg="server config" env="map[CUDA_VISIBLE_DEVICES: GGML_VK_VISIBL
E_DEVICES: GPU_DEVICE_ORDINAL: HIP_VISIBLE_DEVICES: HSA_OVERRIDE_GFX_VERSION: HTTPS_PROXY: HTTP_PROXY: LLAMA_ARG_FIT: LLAMA_ARG_FIT_T
ARGET: NO_PROXY: OLLAMA_CONTEXT_LENGTH:0 OLLAMA_DEBUG:INFO OLLAMA_DEBUG_LOG_REQUESTS:false OLLAMA_EDITOR: OLLAMA_FLASH_ATTENTION:fals
e OLLAMA_GO_TEMPLATE:true OLLAMA_GPU_OVERHEAD:0 OLLAMA_HOST:http://127.0.0.1:11434 OLLAMA_IGPU_ENABLE: OLLAMA_KEEP_ALIVE:5m0s OLLAMA_
KV_CACHE_TYPE: OLLAMA_LLM_LIBRARY: OLLAMA_LOAD_TIMEOUT:5m0s OLLAMA_MAX_LOADED_MODELS:0 OLLAMA_MAX_QUEUE:512 OLLAMA_MAX_TRANSFER_STREA
MS:4 OLLAMA_MODELS:C:\\Users\\Sanjana CA\\.ollama\\models OLLAMA_NOHISTORY:false OLLAMA_NOPRUNE:false OLLAMA_NO_CLOUD:false OLLAMA_NU
M_PARALLEL:1 OLLAMA_ORIGINS:[http://localhost https://localhost http://localhost:* https://localhost:* http://127.0.0.1 https://127.0
.0.1 http://127.0.0.1:* https://127.0.0.1:* http://0.0.0.0 https://0.0.0.0 http://0.0.0.0:* https://0.0.0.0:* app://* file://* tauri:
//* vscode-webview://* vscode-file://*] OLLAMA_REMOTES:[ollama.com] OLLAMA_SCHED_SPREAD:false OLLAMA_VULKAN:true ROCR_VISIBLE_DEVICES
:]"
time=2026-08-10T16:14:38.703+05:30 level=INFO source=routes.go:1935 msg="Ollama cloud disabled: false"
time=2026-08-10T16:14:38.800+05:30 level=INFO source=images.go:883 msg="total blobs: 11"
time=2026-08-10T16:14:38.805+05:30 level=INFO source=images.go:890 msg="total unused blobs removed: 0"
time=2026-08-10T16:14:38.821+05:30 level=INFO source=routes.go:1990 msg="Listening on 127.0.0.1:11434 (version 0.32.6)"
time=2026-08-10T16:14:38.844+05:30 level=INFO source=runner.go:60 msg="discovering available GPUs..."
time=2026-08-10T16:14:39.406+05:30 level=INFO source=model_list_cache.go:112 msg="model list cache hydration complete" models=2 failu
res=0 elapsed=584.1ms
time=2026-08-10T16:14:39.647+05:30 level=INFO source=model_recommendations.go:177 msg="model recommendations cache sleep scheduled" w
ait=3h16m49.79700552s consecutive_failures=0
time=2026-08-10T16:15:05.028+05:30 level=INFO source=runner.go:405 msg="dropping integrated GPU; to enable, set OLLAMA_IGPU_ENABLE=1"
 id=0 library=Vulkan compute=0.0 name=Vulkan0 description="AMD Radeon(TM) Graphics" pci_id=""
time=2026-08-10T16:15:05.028+05:30 level=INFO source=types.go:50 msg="inference compute" id=cpu library=cpu compute="" name=cpu descr
iption=cpu libdirs=ollama driver="" pci_id="" type="" total="15.3 GiB" available="5.2 GiB"
time=2026-08-10T16:15:05.028+05:30 level=INFO source=routes.go:2040 msg="vram-based default context" total_vram="0 B" default_num_ctx
=4096
[GIN] 2026/08/10 - 16:16:00 | 200 |            0s |       127.0.0.1 | GET      "/api/version"

C:\Users\Sanjana CA>
```

ENG IN 16:57 10-08-2026

```
C:\WINDOWS\system32\cmd.e   Windows PowerShell   Windows PowerShell

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Sanjana CA> pip install requests
Requirement already satisfied: requests in .\AppData\Local\Programs\Python\Python314\Lib\site-packages (2.34.2)
Requirement already satisfied: charset_normalizer<4,>=2 in .\AppData\Local\Programs\Python\Python314\Lib\site-packages (from requests
) (3.4.9)
Requirement already satisfied: idna<4,>=2.5 in .\AppData\Local\Programs\Python\Python314\Lib\site-packages (from requests) (3.18)
Requirement already satisfied: urllib3<3,>=1.26 in .\AppData\Local\Programs\Python\Python314\Lib\site-packages (from requests) (2.7.0
)
Requirement already satisfied: certifi>=2023.5.7 in .\AppData\Local\Programs\Python\Python314\Lib\site-packages (from requests) (2026
.6.17)

[notice] A new release of pip is available: 26.1.2 -> 26.2
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\Sanjana CA>
```

ENG IN 17:08 10-08-2026

C:\WINDOWS\system32\cmd.e | Windows PowerShell | Windows PowerShell

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Sanjana CA> ollama pull llama3.2:3b
pulling manifest
pulling dde5aa3fc5ff: 100%                2.0 GB
pulling 966de95ca8a6: 100%                1.4 KB
pulling fcc5a6bec9da: 100%                7.7 KB
pulling a70ff7e570d9: 100%                6.0 KB
pulling 56bb8bd477a5: 100%                 96 B
pulling 34bb5ab01051: 100%                561 B
verifying sha256 digest
writing manifest
success
PS C:\Users\Sanjana CA> ollama run llama3.2:3b
>>> What is machine learning?
Machine learning (ML) is a subset of artificial
intelligence (AI) that involves training algorithms to
learn from data, make predictions, and improve their
performance over time. In other words, ML enables machines
to automatically improve their accuracy on a specific task
without being explicitly programmed.

The basic idea behind machine learning is to provide the
algorithm with a large dataset, which it uses to identify
patterns, relationships, and trends. The algorithm then
learns from this data by adjusting its parameters and
improving its predictions based on feedback or error
analysis.

Machine learning can be categorized into several types,
including:

1. **Supervised Learning**: This type of ML involves
training the algorithm on labeled data, where the correct
output is already known. The goal is to learn a mapping between inputs (features) and outputs (labels), so the algorithm can
make accurate predictions on new, unseen data.
```

ENG IN 17:09 10-08-2026

C:\WINDOWS\system32\cmd.e | Windows PowerShell | Windows PowerShell

```
1. **Supervised Learning**: This type of ML involves
training the algorithm on labeled data, where the correct
output is already known. The goal is to learn a mapping between inputs (features) and outputs (labels), so the algorithm can
make accurate predictions on new, unseen data.
2. **Unsupervised Learning**: In this type of ML, the algorithm is trained on unlabeled data, and it must identify patterns or
structure within the data itself. The goal is to discover hidden relationships or groupings in the data that may not be
immediately apparent.
3. **Reinforcement Learning**: This type of ML involves training an algorithm through trial and error by interacting with an
environment. The algorithm receives feedback in the form of rewards or penalties, which it uses to adjust its behavior and
improve its performance.

Machine learning has numerous applications across various industries, including:

1. **Image Recognition**: Facial recognition, object detection, image classification
2. **Natural Language Processing (NLP)**: Text analysis, sentiment analysis, language translation
3. **Predictive Analytics**: Predicting customer behavior, forecasting demand, identifying risk factors
4. **Recommendation Systems**: Suggesting products or services based on user preferences
5. **Robotics and Autonomous Vehicles**: Improving navigation, control, and decision-making in robots and self-driving cars

Machine learning has many benefits, including:

1. **Improved accuracy**: By leveraging large datasets and complex algorithms, ML can provide highly accurate results.
2. **Increased efficiency**: Machine learning can automate tasks, reduce manual labor, and improve productivity.
3. **Enhanced decision-making**: By providing insights and predictions based on data analysis, ML can support informed
decision-making.

However, machine learning also has some limitations and challenges, including:

1. **Data quality issues**: Poor data quality or lack of relevant data can significantly impact the accuracy of ML models.
2. **Overfitting**: Models that are too complex or overfitted to a specific dataset may not generalize well to new data.
3. **Explainability**: It can be challenging to interpret and understand the decisions made by machine learning models.

Overall, machine learning is a powerful technology with many applications and benefits, but it requires careful consideration
of its limitations and challenges to achieve optimal results.

>>> Send a message (/? for help)
```

ENG IN 17:09 10-08-2026

```
Windows PowerShell
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PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship> python W5D5_local_qa_bot.py
W5D5 - Ollama Local LLM Inference
==================================================

Prompt 1: What is machine learning?
--------------------------------------------------
Response:
Machine learning (ML) is a type of artificial intelligence (AI) that allows computers to learn from data without being explicitly pro
grammed.

Imagine you're teaching a child to recognize pictures of dogs and cats. You show them many images and tell them "this is a dog" or "t
his is a cat". Over time, they learn to identify the animals on their own.

Machine learning works in a similar way. A computer program is trained on a large dataset of examples (like your child with the pictu
res). As it learns from these examples, it becomes better at recognizing patterns and making predictions or decisions.

There are different types of machine learning, including:

1. Supervised learning (e.g., image recognition)
2. Unsupervised learning (e.g., grouping similar items together)
3. Reinforcement learning (e.g., teaching a robot to play a game)

Machine learning is used in many applications, such as:

* Image and speech recognition
* Predicting customer behavior
* Self-driving cars

Does that make sense?

Prompt 2: What is the difference between AI and machine learning?
--------------------------------------------------
Response:
AI (Artificial Intelligence) and Machine Learning (ML) are related but not exactly the same.
```

```
Prompt 2: What is the difference between AI and machine learning?
--------------------------------------------------
Response:
AI (Artificial Intelligence) and Machine Learning (ML) are related but not exactly the same.

**Artificial Intelligence (AI)**:
AI refers to the broader field of research and development aimed at creating intelligent machines that can perform tasks that typical
ly require human intelligence, such as:

* Reasoning
* Problem-solving
* Learning
* Perception

Think of AI like a smart robot that can think and act like humans.

**Machine Learning (ML)**:
ML is a subset of AI that focuses specifically on developing algorithms and statistical models to enable machines to learn from data
without being explicitly programmed.

In other words, ML is a way for computers to improve their performance on a task by automatically learning from experience and feedba
ck, rather than relying on hand-coding.

To illustrate the difference:

* AI is like creating a smart robot that can think and act like humans.
* Machine Learning is like teaching a smart robot how to learn from data and make decisions without being explicitly programmed.

Does that help clarify the difference?

Prompt 3: What is a Python list?
--------------------------------------------------
Response:
In Python, a **list** is an ordered collection of items that can be of any data type, including strings, integers, floats, and other
lists.
```

```
Think of it like a box where you can store multiple values. You can add, remove, or change individual values in the list, but the ord
er remains the same.

Here's an example:

```python
my_list = [1, 2, "hello", 4.5]
print(my_list)  # Output: [1, 2, 'hello', 4.5]

# Adding a new value to the end of the list:
my_list.append(6)
print(my_list)  # Output: [1, 2, 'hello', 4.5, 6]
```

Python lists are useful for storing and manipulating collections of data in a program.

Prompt 4: What is an API?

Response:
An API (Application Programming Interface) is like a messenger between two systems. It allows different systems to talk to each other
 and exchange information.

Think of it like ordering food at a restaurant:

* You (the user) are the customer, and you want to order food.
* The restaurant (system) has a menu with options that you can choose from.
* The chef (system) cooks your food according to your order.
* The waiter (API) takes your order, delivers it to the kitchen, and receives the cooked food back.

In this example:

* You are using the API (ordering food).
* The restaurant system is providing the menu (information) to you.
* The chef system is processing your order and sending the response back to the API (delivered cooked food).

APIs help different systems communicate with each other, making it easier to build applications that can interact with other services
```

```
APIs help different systems communicate with each other, making it easier to build applications that can interact with other services
 or data sources.

Prompt 5: Why are vector databases useful for AI applications?
--------------------------------------------------
Response:
Vector databases are useful for AI applications because they enable fast similarity searches between vectors ( numerical representati
ons of data) in high-dimensional spaces.

Think of it like a library where you have many books with different keywords. In a traditional database, you'd need to search through
 all the books one by one to find matches. But with vector databases, you can use similarity measures like cosine similarity or dista
nce metrics to quickly find similar "books" (vectors) in a massive space.

This makes them particularly useful for:

1. **Embedding models**: AI models that learn to represent data as vectors (e.g., word embeddings).
2. **Recommendation systems**: Finding similar items based on user behavior and preferences.
3. **Anomaly detection**: Identifying unusual patterns or outliers in large datasets.

By leveraging vector databases, you can improve the efficiency and accuracy of these AI applications!
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship> ollama pull qwen2.5:3b
pulling manifest
pulling 5ee4f07cdb9b: 100%                                                          1.9 GB
pulling 66b9ea09bd5b: 100%                                                           68 B
pulling eb4402837c78: 100%                                                          1.5 KB
pulling b5c0e5cf74cf: 100%                                                          7.4 KB
pulling 161ddde4c9cd: 100%                                                           487 B
verifying sha256 digest
writing manifest
success
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship> ollama list
NAME          ID              SIZE      MODIFIED
qwen2.5:3b    357c53fb659c    1.9 GB    12 seconds ago
llama3.2:3b   a80c4f17acd5    2.0 GB    18 minutes ago
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship> python W5D5_model_comparison.py
W5D5 - Model Comparison
```

```
W5D5 - Model Comparison
======================================================================

QUESTION 1: What is machine learning?
======================================================================

MODEL: llama3.2:3b
----------------------------------------------------------------------
Machine learning (ML) is a way to teach computers to learn from data without being explicitly programmed. It's a type of artificial i
ntelligence (AI) that allows computers to:

1. Look at data
2. Identify patterns
3. Make predictions or decisions based on that data

Think of it like this: Imagine you show a child many pictures of dogs and cats. Over time, they can learn to recognize what a dog and
 cat look like, even if they haven't seen them before. Machine learning does something similar with computers and data!

MODEL: qwen2.5:3b
----------------------------------------------------------------------
Machine learning is a type of artificial intelligence that allows software to get better at predicting outcomes or making decisions w
ithout being explicitly programmed. It involves training algorithms with data so they can make predictions or take actions based on p
atterns it has learned from the input data.

QUESTION 2: What is an API? Give a simple example.
======================================================================

MODEL: llama3.2:3b
----------------------------------------------------------------------
**What is an API?**

An Application Programming Interface (API) is like a messenger between two systems.

Imagine you're at a restaurant and you want to order food. The waiter is like the API. You give them your order, and they take it to
the kitchen staff, who prepare your food. Then, the waiter brings the prepared food back to you.

In this example:
```

```
In this example:

* Your request (order) is like the data you want to send to another system.
* The API (waiter) receives your request, processes it, and sends it to the next system (kitchen).
* The next system (kitchen) processes the request and sends the response back through the API (waiter).

**Simple Example:**

Suppose you're building a weather app that shows the current temperature in New York City. You want to display this data on your app.

Here's how it works:

1. Your app makes an API call to a weather service like OpenWeatherMap.
2. The API receives your request (get current temperature for NYC) and sends it to OpenWeatherMap.
3. OpenWeatherMap processes the request, retrieves the current temperature, and returns the data back to the API.
4. The API sends the response (current temperature in NYC) back to your app.

Your app then displays the current temperature in NYC to the user!

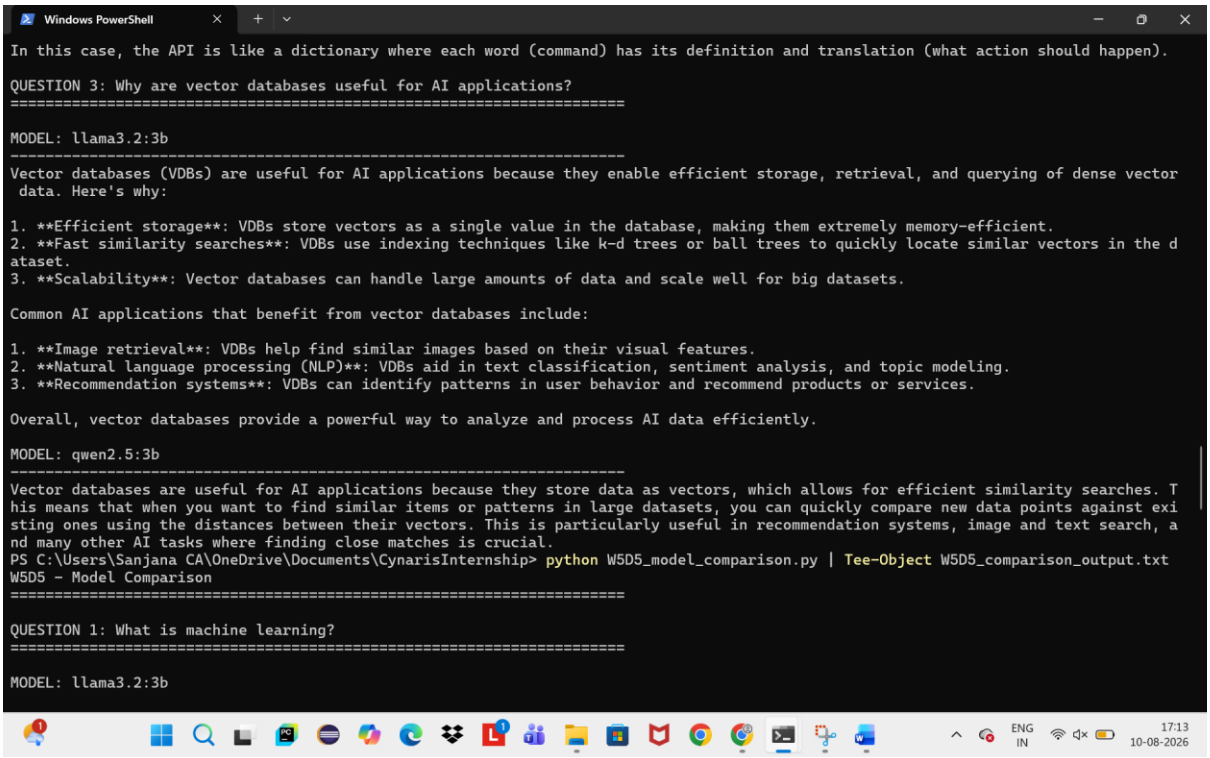
In this example, the API acts as a messenger between your weather app and OpenWeatherMap, allowing you to retrieve the data you need
without having to build it yourself.

MODEL: qwen2.5:3b
----------------------------------------------------------------------
An API stands for Application Programming Interface. It's like a set of rules that different pieces of software use to talk to each o
ther and exchange information.

A Simple Example:
Imagine you have a toy robot, and you want it to play music when you give it commands through your phone app (the user interface). Th
e communication between the phone app (software) and the toy robot (another piece of software) uses an API.

The phone app follows certain rules defined by the API so that it can tell the toy robot what songs to play, which volumes to use, et
c. Without these rules or an API, it would be much harder for one device/software to understand instructions from another.

In this case, the API is like a dictionary where each word (command) has its definition and translation (what action should happen).
```



```
Windows PowerShell
In this case, the API is like a dictionary where each word (command) has its definition and translation (what action should happen).

QUESTION 3: Why are vector databases useful for AI applications?
======================================================================

MODEL: llama3.2:3b
----------------------------------------------------------------------
Vector databases (VDBs) are useful for AI applications because they enable efficient storage, retrieval, and querying of dense vector
 data. Here's why:

1. **Efficient storage**: VDBs store vectors as a single value in the database, making them extremely memory-efficient.
2. **Fast similarity searches**: VDBs use indexing techniques like k-d trees or ball trees to quickly locate similar vectors in the d
ataset.
3. **Scalability**: Vector databases can handle large amounts of data and scale well for big datasets.

Common AI applications that benefit from vector databases include:

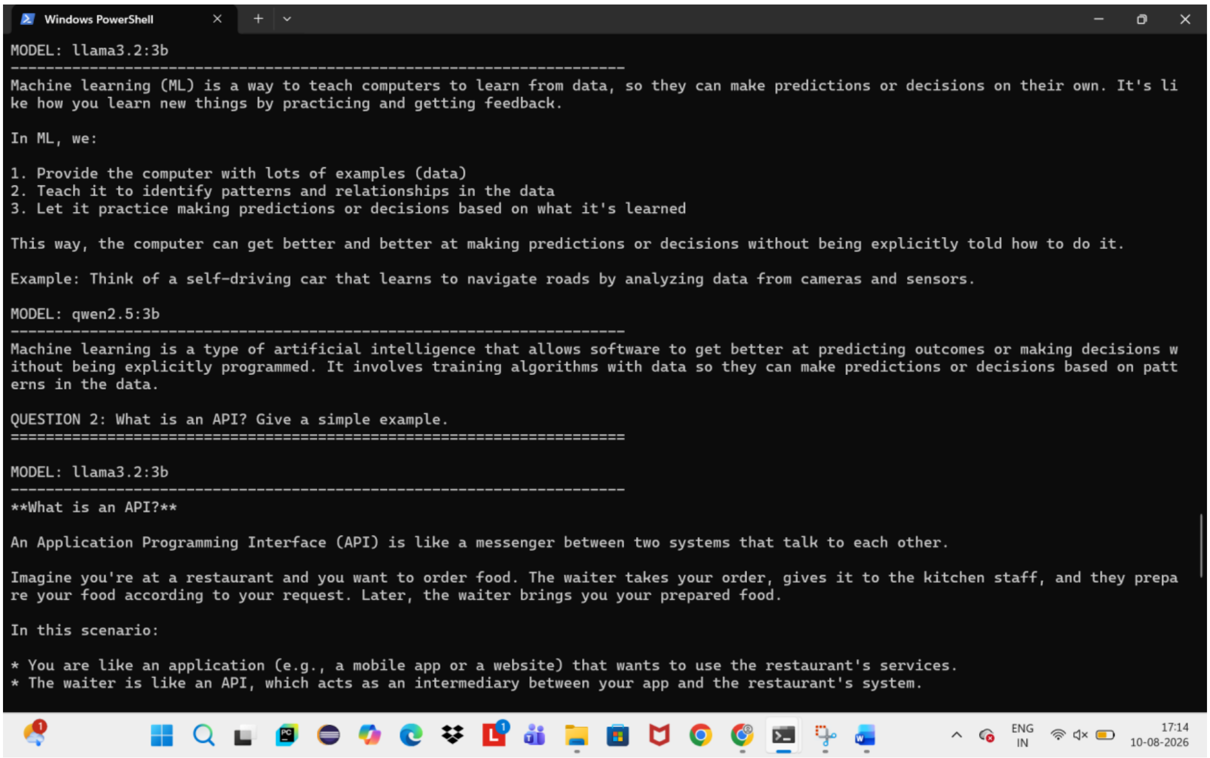
1. **Image retrieval**: VDBs help find similar images based on their visual features.
2. **Natural language processing (NLP)**: VDBs aid in text classification, sentiment analysis, and topic modeling.
3. **Recommendation systems**: VDBs can identify patterns in user behavior and recommend products or services.

Overall, vector databases provide a powerful way to analyze and process AI data efficiently.

MODEL: qwen2.5:3b
----------------------------------------------------------------------
Vector databases are useful for AI applications because they store data as vectors, which allows for efficient similarity searches. T
his means that when you want to find similar items or patterns in large datasets, you can quickly compare new data points against exi
sting ones using the distances between their vectors. This is particularly useful in recommendation systems, image and text search, a
nd many other AI tasks where finding close matches is crucial.
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship> python W5D5_model_comparison.py | Tee-Object W5D5_comparison_output.txt
W5D5 - Model Comparison
======================================================================

QUESTION 1: What is machine learning?
======================================================================

MODEL: llama3.2:3b
ENG IN 17:13 10-08-2026
```



```
Windows PowerShell
MODEL: llama3.2:3b
----------------------------------------------------------------------
Machine learning (ML) is a way to teach computers to learn from data, so they can make predictions or decisions on their own. It's li
ke how you learn new things by practicing and getting feedback.

In ML, we:

1. Provide the computer with lots of examples (data)
2. Teach it to identify patterns and relationships in the data
3. Let it practice making predictions or decisions based on what it's learned

This way, the computer can get better and better at making predictions or decisions without being explicitly told how to do it.

Example: Think of a self-driving car that learns to navigate roads by analyzing data from cameras and sensors.

MODEL: qwen2.5:3b
----------------------------------------------------------------------
Machine learning is a type of artificial intelligence that allows software to get better at predicting outcomes or making decisions w
ithout being explicitly programmed. It involves training algorithms with data so they can make predictions or decisions based on patt
erns in the data.

QUESTION 2: What is an API? Give a simple example.
======================================================================

MODEL: llama3.2:3b
----------------------------------------------------------------------
**What is an API?**

An Application Programming Interface (API) is like a messenger between two systems that talk to each other.

Imagine you're at a restaurant and you want to order food. The waiter takes your order, gives it to the kitchen staff, and they prepa
re your food according to your request. Later, the waiter brings you your prepared food.

In this scenario:

* You are like an application (e.g., a mobile app or a website) that wants to use the restaurant's services.
* The waiter is like an API, which acts as an intermediary between your app and the restaurant's system.
ENG IN 17:14 10-08-2026
```

Windows PowerShell

```
**Simple Example:**

Suppose you have a weather app that needs to fetch current weather data from a weather website. You can use an API to communicate wit
h this website like this:

1. Your app (weather app) makes a request to the weather API.
2. The API sends your request to the weather website's server.
3. The weather website's server responds with the current weather data.
4. The API receives the response and sends it back to your app.

This way, your app can access the weather data without having to know the details of how the weather website works.

That's a basic overview of APIs!

MODEL: qwen2.5:3b
----------------------------------------------------------------------
An API, or Application Programming Interface, is like a set of rules that allows different software applications to communicate with
each other. It's how different parts of a program talk to each other.

A Simple Example:
Imagine you have a toy box (which could be the website) and inside it are several toys (like JavaScript files). One toy can give inst
ructions on how to play with another toy directly without much explanation or setup needed, just like an API. So if you want to make
your teddy bear light up using a magic flashlight (JavaScript), you would use the API provided by the flashlight app that tells the t
eddy bear how to respond.

In this analogy:
- The toy box = Website
- The toys inside = Code or JavaScript files
- The instructions for playing = API

The simplest way to understand an API is through HTTP requests. For example, if you want to use a weather website's current temperatu
re information in your app (the API), you send a request asking the weather site for that data and get back a response with the tempe
rature value or JSON object containing that information.

Remember, APIs are not just about toy boxes and instructions; they can also involve complex interactions depending on what kind of ap
plications and services are involved.
```

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Windows PowerShell


```
eddy bear how to respond.

In this analogy:
- The toy box = Website
- The toys inside = Code or JavaScript files
- The instructions for playing = API

The simplest way to understand an API is through HTTP requests. For example, if you want to use a weather website's current temperatu
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Remember, APIs are not just about toy boxes and instructions; they can also involve complex interactions depending on what kind of ap
plications and services are involved.
```


```
QUESTION 3: Why are vector databases useful for AI applications?
======================================================================

MODEL: llama3.2:3b
----------------------------------------------------------------------
Vector databases are useful for AI applications because they:

1. **Efficiently store and query dense vectors**: Vector databases are optimized to handle large amounts of data stored as vectors (m
athematical representations of objects in a multidimensional space).
2. **Enable fast similarity searches**: By using efficient algorithms like approximate nearest neighbors, vector databases can quickl
y find similar vectors, which is useful for tasks like image recognition, recommendation systems, and natural language processing.
3. **Simplify the integration of AI models**: Vector databases provide a standardized way to represent and store data used by various
 AI models, making it easier to integrate and deploy these models in applications.

Overall, vector databases help accelerate the development and deployment of AI applications by providing fast, efficient, and scalabl
e storage and querying capabilities for dense vector data.

MODEL: qwen2.5:3b
----------------------------------------------------------------------
Vector databases are useful for AI applications because they allow storing and querying data using vectors, which can represent compl
ex information in a way that machines understand. This makes it easier to find similar items or patterns within large datasets, enabl
ing better recommendations, search results, and intelligent systems like chatbots and recommendation engines.
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship>
```

ENG IN 17:15 10-08-2026